

Failed Login Investigation:

In this , we are going to investigate the a failed login and find out whether the user is successful to get access

```
Cmd -- index="main" event_id="4625" | table timestamp, user, workstation, source_ip, failure_reason | sort timestamp
```

The screenshot shows a Splunk search interface with the query `index="main" event_id="4625" | table timestamp, user, workstation, source_ip, failure_reason | sort timestamp`. The results are displayed in a table with 5 columns: timestamp, user, workstation, source_ip, and failure_reason. The table is sorted by timestamp. The first 10 rows show failed login attempts for users mwilson, amartin, mbaker, and fclark. The failure reason for all is "Bad password".

timestamp	user	workstation	source_ip	failure_reason
2024-03-01 08:06:36	mwilson	WIN-HR-002	10.10.1.102	Bad password
2024-03-01 08:06:36	mwilson	WIN-HR-002	10.10.1.102	Bad password
2024-03-01 08:06:36	mwilson	WIN-HR-002	10.10.1.102	Bad password
2024-03-01 08:27:06	amartin	WIN-FIN-002	10.10.2.102	Bad password
2024-03-01 08:27:06	amartin	WIN-FIN-002	10.10.2.102	Bad password
2024-03-01 08:27:06	amartin	WIN-FIN-002	10.10.2.102	Bad password
2024-03-01 08:47:30	mbaker	WIN-SOC-001	10.10.4.101	Bad password
2024-03-01 08:47:30	mbaker	WIN-SOC-001	10.10.4.101	Bad password
2024-03-01 08:47:30	mbaker	WIN-SOC-001	10.10.4.101	Bad password
2024-03-01 09:02:02	fclark	WIN-FIN-006	10.10.2.106	Bad password
2024-03-01 09:02:02	fclark	WIN-FIN-006	10.10.2.106	Bad password
2024-03-01 09:02:02	fclark	WIN-FIN-006	10.10.2.106	Bad password
2024-03-01 09:39:16	fclark	WIN-FIN-006	10.10.2.106	Bad password

Count how many times does each user failed for login:

```
Cmd – index="main" event_id="4625" | stats count by user | sort -count
```

cmd

The screenshot shows a Splunk search interface with the query `index="main" event_id="4625" | stats count by user | sort -count`. The results are displayed in a table with 2 columns: user and count. The table is sorted by count in descending order. The user rthompson has the highest count of 540 failed login attempts.

user	count
rthompson	540
qkelly	75
wevans	72
mbaker	69
vparker	69
ucampbell	66
kgreen	63
pbrooks	63
sturner	63
rroberts	60
sprice	60
asanchez	57
dreed	57

Now we have user named rthompson who had logined attempts nearly 540

We will penetrate through him and find out whether he is successful in login or not

Cmd--- index="main" event_id="4625" |stats count by source_ip | sort – count

New Search Save As Create Table View Close

1 index="main" event_id="4625" |stats count by source_ip | sort – count All time Q

✓ 2,925 events (before 9/15/26 9:02:29.000 AM) No Event Sampling Job II III ↶ ↷ ⬇ Smart Mode

Events Patterns **Statistics (52)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 3 Next >

source_ip	count
203.0.113.50	471
10.10.9.102	75
10.10.5.106	72
10.10.2.101	69
10.10.4.101	69
10.10.5.105	66
10.10.3.104	63
10.10.5.102	63
10.10.5.104	63
10.10.9.101	63

Using the ip 203.0.113.50 , we will get the complete details

Cmd -- index="main" source_ip="203.0.113.50" event_id="4625"

| table timestamp, user, failure_reason, source_ip | sort timestamp

1 index="main" source_ip="203.0.113.50" event_id="4625"

2 | table timestamp, user, failure_reason, source_ip | sort timestamp All time Q

✓ 471 events (before 9/15/26 9:05:29.000 AM) No Event Sampling Job II III ↶ ↷ ⬇ Smart Mode

Events Patterns **Statistics (471)** Visualization

Show: 20 Per Page Format Preview: On < Prev 1 2 3 4 5 6 7 8 ... Next >

timestamp	user	failure_reason	source_ip
2024-03-15 02:15:00	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:00	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:00	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:08	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:08	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:08	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:16	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:16	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:16	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:24	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:24	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:24	rthompson	Bad password	203.0.113.50
2024-03-15 02:15:32	rthompson	Bad password	203.0.113.50

By clearly observing the table , we can see that the attempt to login in time is only difference in 8 seconds that to in a span of 15 minutes , looks suspicious

Now we will see the use successfully get the password or not

Cmd -- index="main" source_ip="203.0.113.50" | stats count by event_id,event_name,user

New Search Save As Create Table View Close

1 `index="main" source_ip="203.0.113.50"`
 2 `| stats count by event_id,event_name,user` All time Q

✓ 480 events (before 9/15/26 9:09:39.000 AM) No Event Sampling Job || → ⌵ ⌴ ⌵ Smart Mode

Events Patterns Statistics (2) Visualization

Show: 20 Per Page Format Preview: On

event_id	event_name	user	count
4624	SuccessfulLogon	rthompson	9
4625	FailedLogon	rthompson	471

The user successfully login 9 times

Now we will see at which time he successfully logged

Cmd- `index="main" source_ip="203.0.113.50" event_id=4624 event_name=SuccessfulLogon user=rthompson`

New Search Save As Create Table View Close

1 `index="main" source_ip="203.0.113.50" event_id=4624 event_name=SuccessfulLogon user=rthompson` All time Q

✓ 9 events (before 9/15/26 9:14:06.000 AM) No Event Sampling Job || → ⌵ ⌴ ⌵ Smart Mode

Events (9) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 100 milliseconds per column

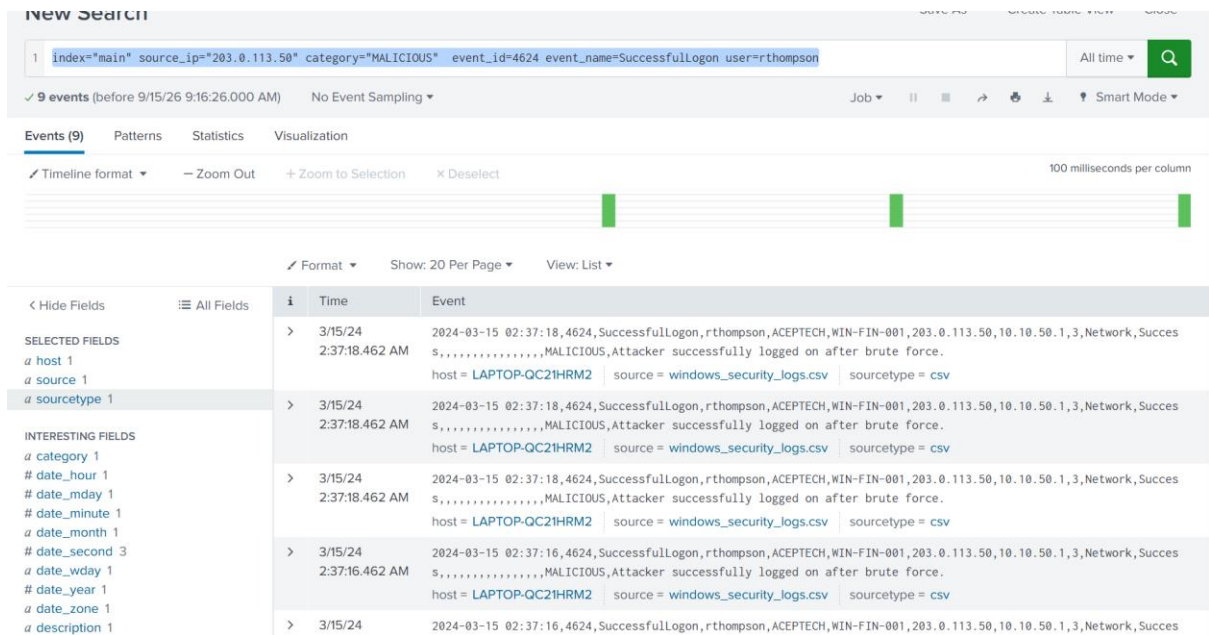
Format Show: 20 Per Page View: List

	Time	Event
>	3/15/24 2:37:18.462 AM	2024-03-15 02:37:18,4624,SuccessfulLogon,rthompson,ACEPTECH,WIN-FIN-001,203.0.113.50,10.10.50.1,3,Network,Success,,,,,,MALICIOUS,Attacker successfully logged on after brute force. host = LAPTOP-QC21HRM2 source = windows_security_logs.csv sourcetype = csv
>	3/15/24 2:37:18.462 AM	2024-03-15 02:37:18,4624,SuccessfulLogon,rthompson,ACEPTECH,WIN-FIN-001,203.0.113.50,10.10.50.1,3,Network,Success,,,,,,MALICIOUS,Attacker successfully logged on after brute force. host = LAPTOP-QC21HRM2 source = windows_security_logs.csv sourcetype = csv
>	3/15/24 2:37:18.462 AM	2024-03-15 02:37:18,4624,SuccessfulLogon,rthompson,ACEPTECH,WIN-FIN-001,203.0.113.50,10.10.50.1,3,Network,Success,,,,,,MALICIOUS,Attacker successfully logged on after brute force. host = LAPTOP-QC21HRM2 source = windows_security_logs.csv sourcetype = csv
>	3/15/24 2:37:16.462 AM	2024-03-15 02:37:16,4624,SuccessfulLogon,rthompson,ACEPTECH,WIN-FIN-001,203.0.113.50,10.10.50.1,3,Network,Success,,,,,,MALICIOUS,Attacker successfully logged on after brute force. host = LAPTOP-QC21HRM2 source = windows_security_logs.csv sourcetype = csv

so the user successfully logged at 2.37 am --- this is malicious

By checking with malicious category, we can find whether this attempt is really malicious or not

Cmd -- `index="main" source_ip="203.0.113.50" category="MALICIOUS" event_id=4624 event_name=SuccessfulLogon user=rthompson`



By all these evidence , we can indicate that the user performed a bruteforce attack and successfully gained access to the passwords